Supplementary Material for "Testing Limited Overlap"

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Abstract

This supplementary material presents additional mathematical proofs and details that are omitted from the main text. Section E contains the proof of Theorem 3 along with the associated lemmas and their proofs. Section F contains additional details for Remarks 1 and 2. All the notations $O_p(\cdot)$ and $o_p(\cdot)$ are based on $n \to \infty$.

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Appendix

E Proof of Theorem 3

To prove Theorem 3, we first establish some auxiliary lemmas.

Lemma E.1. *If Assumption 4 (i–iv) is satisfied, then* $\|\hat{\psi} - \psi_0\|_1 = o_p(1)$ *and* $\|\hat{\psi} - \psi_0\|_2 = o_p(1)$.

Proof. The claim in this lemma follows from Bierens (2014, Theorem 4.2) with the metric d. We thus check his Assumptions 4.1 and 4.2 with d. Assumption 4.1 (a) follows from our Assumption 4 (i). Assumption 4.1 (b) follows from our definitions of Ψ and d. Assumption 4.1 (c) follows from our definition of g. Assumption 4.1 (d) follows from Lemma 4.1 of Bierens (2014). Assumption 4.1 (e) holds trivially. Assumption 4.1 (f) follows from Lemma 2.1 of Bierens (2014) under our Assumption 4 (ii–iv). Assumption 4.1 (g) (h) follows from our definitions of Ψ and Ψ_n. Assumption 4.1 (i) holds with $\psi_n = (\beta_0', \Xi_{\ell_n - p}\delta_0')'$ – see Section 4.2.2 of Bierens (2014). Assumption 4.1 (j) follows from our Assumption 4 (iv) and that G is a logistic link. Assumption 4.2 follows from our definitions of Ψ and Ψ_n. □

Lemma E.2. *If Assumption 4 is satisfied, then* $\|\hat{\beta} - \beta_0\|_2 = O_p(n^{-1/2})$.

Proof. The claim in this lemma follows from Bierens (2014, Theorem 6.1) along with our definition of the metric $d_{(2)}$. We thus check his Assumptions 6.1–6.8. Assumptions 6.1 (a) and 6.2 follow from the same argument as in the proof of Lemma E.1 under our Assumption 4 (i–iv). Assumption 6.1 (b) (c) (d) follows from our Assumption 4 (iv–v). Assumptions 6.3, 6.4 and 6.5 follow by Lemma 7.1 of Bierens (2014) from our definitions of Ψ and Ψ_n under our Assumption 4 (ii-v). Assumption 6.6 follows from our Assumption 4 (v). Assumptions 6.7 and 6.8 follow from our Assumption 4 (v). □

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Lemma E.3. *Assume Assumption 4 holds. Let c_n be a vanishing sequence satisfying*

$$\max_{1\leq i\leq n}\sup_{\|\beta-\beta_0\|\leq\frac{c}{\sqrt{n}}}\left\|\frac{G(X_i'\beta_0)}{G(X_i'\beta)^2}\frac{\partial G(X_i'\beta)}{\partial\beta}\right\| = O_p\left(\sqrt{n}c_n\right)$$

for all $c>0$. Then

$$\max_{1\leq i\leq n}\left|\frac{G(X_i'\beta_0)}{G(X_i'\hat{\beta})}-1\right| = O_p(c_n).$$

Proof. This lemma can be established using the same proof strategy as in Theorem 2. □

Lemma E.4. *Suppose Assumption 4 holds. If, in addition, that H is continuously differentiable with uniformly bounded derivatives, then*

$$\max_{1\leq i\leq n}\left|\frac{H(G(X_i'\hat{\beta});\delta_0)-H(G(X_i'\beta_0);\delta_0)}{G(X_i'\beta_0)}\right| = o_p(1).$$

Proof. The result follows from Lemma E.3 and the mean value expansion under Assumption 4 (iv) implying that H is continuously differentiable with uniformly bounded derivatives. □

Lemma E.5. *Suppose Assumption 4 holds. Then*

$$\sup_{u\in(0,1)}\left|\frac{H(u;\hat{\delta})-H(u;\delta_0)}{u}\right| = o_p(1).$$

Proof. First, note that the conclusion of Lemma E.1 implies

$$\|\hat{\delta}-\delta_0\|_1 = o_p(1), \tag{E.1}$$

$$\|\hat{\delta}-\delta_0\|_2 = o_p(1). \tag{E.2}$$

Furthermore, by the definition of the parameter space Ψ, we have

$$\sum_{j=1}^{\infty} \delta_{0j}^2 < \infty. \tag{E.3}$$

By triangle inequality and Cauchy-Schwartz inequality,

$$\left|\sum_{j=1}^{\infty}(\hat{\delta}_j^2 - \delta_{0j}^2)\right| \leq \sum_{j=1}^{\infty}(\hat{\delta}_j - \delta_{0j})^2 + 2\left(\sum_{j=1}^{\infty}(\hat{\delta}_j - \delta_{0j})^2\right)^{1/2}\left(\sum_{j=1}^{\infty}\delta_{0j}^2\right)^{1/2} = o_p(1)$$

and

$$\begin{aligned}\left|\sum_{j=2}^{\infty}\sum_{m=1}^{j-1}(\hat{\delta}_j\hat{\delta}_m - \delta_{0j}\delta_{0m})\right| &= \left(\sum_{j=2}^{\infty}(\hat{\delta}_j - \delta_{0j})^2\right)^{1/2}\left(\sum_{m=1}^{j-1}(\hat{\delta}_m - \delta_{0m})^2\right)^{1/2} \\ &\quad + \left(\sum_{j=2}^{\infty}\delta_{0j}^2\right)^{1/2}\left(\sum_{m=1}^{j-1}(\hat{\delta}_m - \delta_{0m})^2\right)^{1/2} \\ &\quad + \left(\sum_{j=2}^{\infty}(\hat{\delta}_j - \delta_{0j})^2\right)^{1/2}\left(\sum_{m=1}^{j-1}\delta_{0m}^2\right)^{1/2} = o_p(1)\end{aligned}$$

by (E.2) and (E.3). Therefore,

$$\begin{aligned}&\sum_{j=1}^{\infty}\left(\frac{\hat{\delta}_j}{1+\sum_{i=1}^{\infty}\hat{\delta}_i^2} - \frac{\delta_{0j}}{1+\sum_{i=1}^{\infty}\delta_{0i}^2}\right) \\ &\quad = \frac{(1+\sum_{i=1}^{\infty}\delta_{0i}^2)\sum_{j=1}^{\infty}(\hat{\delta}_j - \delta_{0j}) - \sum_{j=1}^{\infty}\delta_{0j}\sum_{i=1}^{\infty}(\hat{\delta}_i^2 - \delta_{0i}^2)}{(1+\sum_{i=1}^{\infty}\delta_{0i}^2)(1+\sum_{i=1}^{\infty}\delta_{0i}^2 + \sum_{i=1}^{\infty}(\hat{\delta}_i^2 - \delta_{0i}^2))} = o_p(1), \\ &\sum_{j=1}^{\infty}\left(\frac{\hat{\delta}_j^2}{1+\sum_{i=1}^{\infty}\hat{\delta}_i^2} - \frac{\delta_{0j}^2}{1+\sum_{i=1}^{\infty}\delta_{0i}^2}\right) \\ &\quad = \frac{(1+\sum_{i=1}^{\infty}\delta_{0i}^2)\sum_{j=1}^{\infty}(\hat{\delta}_j^2 - \delta_{0j}^2) - \sum_{j=1}^{\infty}\delta_{0j}^2\sum_{i=1}^{\infty}(\hat{\delta}_i^2 - \delta_{0i}^2)}{(1+\sum_{i=1}^{\infty}\delta_{0i}^2)(1+\sum_{i=1}^{\infty}\delta_{0i}^2 + \sum_{i=1}^{\infty}(\hat{\delta}_i^2 - \delta_{0i}^2))} = o_p(1), \qquad \text{and} \\ &\sum_{j=2}^{\infty}\sum_{m=1}^{j-1}\left(\frac{\hat{\delta}_j\hat{\delta}_m}{1+\sum_{i=1}^{\infty}\hat{\delta}_i^2} - \frac{\delta_{0j}\delta_{0m}}{1+\sum_{i=1}^{\infty}\delta_{0i}^2}\right) \\ &\quad = \frac{(1+\sum_{i=1}^{\infty}\delta_{0i}^2)\sum_{j=2}^{\infty}\sum_{m=1}^{j-1}(\hat{\delta}_j\hat{\delta}_m - \delta_{0j}\delta_{0m}) - \sum_{j=2}^{\infty}\sum_{m=1}^{j-1}\delta_{0j}\delta_{0m}\sum_{i=1}^{\infty}(\hat{\delta}_i^2 - \delta_{0i}^2)}{(1+\sum_{i=1}^{\infty}\delta_{0i}^2)(1+\sum_{i=1}^{\infty}\delta_{0i}^2 + \sum_{i=1}^{\infty}(\hat{\delta}_i^2 - \delta_{0i}^2))}\end{aligned}$$

$$= o_p(1)$$

by (E.1), (E.2) and (E.3). Note also that

$$\left| \frac{\sin(j\pi u)}{j\pi u} \right| \leq 1$$

holds for any $j \in \mathbb{Z}$ and $u \in (0,1)$. These equations together imply

$$\sup_{u \in (0,1)} \left| \frac{\Upsilon(u;\hat{\delta}) - \Upsilon(u;\delta_0)}{u\left(1 + \sum_{i=1}^{\infty} \delta_{0i}^2\right)} \right| = o_p(1).$$

From the definition of $H(u;\delta)$, this completes a proof of the lemma. □

With these lemmas, we are now ready to establish Theorem 3.

Proof of Theorem 3. It suffices to prove that

$$\max_{1 \leq i \leq n} \left| \frac{H(G(X_i'\beta_0);\delta_0)}{H(G(X_i'\hat{\beta});\hat{\delta})} - 1 \right| = o_p(1)$$

holds under Assumption 4. First, note that we have

$$\begin{aligned}
&\max_{1 \leq i \leq n} \left| \frac{G(X_i'\hat{\beta}) - G(X_i'\beta_0)}{G(X_i'\beta_0)} \right| = o_p(1), \\
&\max_{1 \leq i \leq n} \left| \frac{H(G(X_i'\hat{\beta});\delta_0) - H(G(X_i'\beta_0);\delta_0)}{G(X_i'\beta_0)} \right| = o_p(1), \qquad \text{and} \\
&\max_{1 \leq i \leq n} \left| \frac{H(G(X_i'\hat{\beta});\hat{\delta}) - H(G(X_i'\hat{\beta});\delta_0)}{G(X_i'\hat{\beta})} \right| = o_p(1)
\end{aligned}$$

by Lemmas E.3, E.4 and E.5, respectively. Together with these uniform convergence, the equality

$$\frac{H(G(X_i'\beta_0);\delta_0)}{H(G(X_i'\hat{\beta});\hat{\delta})} - 1$$

$$
\begin{aligned}
= - \Bigg[& \frac{H(G(X'\hat{\beta});\hat{\delta}) - H(G(X'\hat{\beta});\delta_0)}{G(X'\hat{\beta})} \left(1 + \frac{G(X'\hat{\beta}) - G(X'\beta_0)}{G(X'\beta_0)}\right) \\
& + \frac{H(G(X'\hat{\beta});\delta_0) - H(G(X'\beta_0);\delta_0)}{G(X'\beta_0)} \Bigg] \Bigg/ \Bigg[\frac{H(G(X'\beta_0);\delta_0)}{G(X'\beta_0)} \\
& + \frac{H(G(X'\hat{\beta});\hat{\delta}) - H(G(X'\hat{\beta});\delta_0)}{G(X'\hat{\beta})} \left(1 + \frac{G(X'\hat{\beta}) - G(X'\beta_0)}{G(X'\beta_0)}\right) \\
& + \frac{H(G(X'\hat{\beta});\delta_0) - H(G(X'\beta_0);\delta_0)}{G(X'\beta_0)} \Bigg]
\end{aligned}
$$

and Assumption 4 (iv) that $x \mapsto H(G(x'\beta_0);\delta_0)/G(x'\beta_0)$ is bounded away from zero prove the statement of the theorem. □

F Omitted Details

F.1 Remark 1

Part 1. As the population moment condition is uniquely maximized at β_0, we know that, for example, $\mathbb{E}[D_i \ln\big(G(X_i'\beta_0)\big)] > -\infty$. To show that $\hat{\beta}$ is consistent for β_0, it suffices to provide an integrable envelop function (Lemma 2.4 in Newey and McFadden 1994). Take an arbitrary β in the parameter space $\mathcal{B}$, then $G(X_i'\beta) \geq e^{X_i'\beta}$ up to a multiplicative constant. This implies, again up to some constants that do not depend on β, $\mathbb{E}[D_i \ln\big(G(X_i'\beta)\big)] \geq \mathbb{E}[X_i'\beta] \geq \mathbb{E}[-\|X_i\|]$. This last expectation will be finite since the parameter space is compact and that X_i has a finite $2+\epsilon$ moment. The argument used to bound $\mathbb{E}[(1-D_i)\ln\big(1-(G(X_i'\beta_0))\big)]$ is completely analogous.

Next, we consider the score

$$
s_i(\beta) = \left[\frac{D_i}{G(X_i'\beta)} - \frac{1-D_i}{1-G(X_i'\beta)}\right] G^{(1)}(X_i'\beta) X_i.
$$

To bound its second moment (at the true parameter β_0), it suffices to consider, for example,

$$\begin{aligned}\mathbb{E}\left[\frac{D_i}{G(X_i'\beta_0)^2}G^{(1)}(X_i'\beta_0)^2\|X_i\|^2\right] &= \mathbb{E}\left[\frac{1}{G(X_i'\beta_0)}G^{(1)}(X_i'\beta_0)^2\|X_i\|^2\right] \\ &= \mathbb{E}\left[\Big(1-G(X_i'\beta_0)\Big)\|X_i\|^2\right],\end{aligned}$$

which is finite given our assumption on X_i. To characterize the $\sqrt{n}$-consistency and asymptotic normality of $\hat{\beta}$, what remains is to show the convergence to the Hessian matrix. To make it precise, let

$$\begin{aligned}H_i(\beta) = \Bigg[D_i\frac{G^{(2)}(X_i'\beta) - G(X_i'\beta)\Big(1-G(X_i'\beta)\Big)^2}{G(X_i'\beta)} \\ - (1-D_i)\frac{G^{(2)}(X_i'\beta) + G(X_i'\beta)^2\Big(1-G(X_i'\beta)\Big)}{1-G(X_i'\beta)}\Bigg]X_iX_i',\end{aligned}$$

and we would like to show

$$\mathbb{E}\left[\sup_{\|\beta-\beta_0\|\leq\frac{\epsilon}{\sqrt{n}}}\|H_i(\beta)\|\right] < \infty$$

for all $\epsilon > 0$. This is particularly easy due to the Logit model we consider. To be precise, one has $|G^{(2)}(X_i'\beta)| \leq G(X_i'\beta)$ and $|G^{(2)}(X_i'\beta)| \leq 1 - G(X_i'\beta)$. As a result, $\|H_i(\beta)\| \leq \|X_i\|^2$ up to a multiplicative constant which does not depend on β. This concludes our proof that $\|\hat{\beta} - \beta_0\| = O_p(1/\sqrt{n})$.

Part 2. Now we prove Assumption 3(ii) with an explicit order c_n. To start, note that

$$\left\|\frac{G(X_i'\beta_0)}{G(X_i'\beta)^2}\frac{\partial G(X_i'\beta)}{\partial\beta}\right\| \leq \frac{e^{X_i'\beta_0}}{1+e^{X_i'\beta_0}}\frac{1}{e^{X_i'\beta}}(1+\|X_i\|) \leq e^{X_i'(\beta_0-\beta)}(1+\|X_i\|),$$

which implies that

$$\sup_{\|\beta-\beta_0\|\leq\frac{c}{\sqrt{n}}}\left\|\frac{G(X_i'\beta_0)}{G(X_i'\beta)^2}\frac{\partial G(X_i'\beta)}{\partial\beta}\right\|\leq(1+\|X_i\|)e^{\frac{c}{\sqrt{n}}(1+\|X_i\|)}.$$

As a result,

$$\max_{1\leq i\leq n}\sup_{\|\beta-\beta_0\|\leq\frac{c}{\sqrt{n}}}\left\|\frac{G(X_i'\beta_0)}{G(X_i'\beta)^2}\frac{\partial G(X_i'\beta)}{\partial\beta}\right\|\leq(1+\max_{1\leq i\leq n}\|X_i\|)e^{\frac{c}{\sqrt{n}}(1+\max_{1\leq i\leq n}\|X_i\|)}.$$

As long as $\|X_i\|$ has a finite $2+\epsilon$ moment for some $\epsilon>0$, it will be true that

$$\max_{1\leq i\leq n}\|X_i\|=O_p(n^{\frac{1}{2+\epsilon}})=o_p(\sqrt{n}),$$

and hence

$$\max_{1\leq i\leq n}\sup_{\|\beta-\beta_0\|\leq\frac{c}{\sqrt{n}}}\left\|\frac{G(X_i'\beta_0)}{G(X_i'\beta)^2}\frac{\partial G(X_i'\beta)}{\partial\beta}\right\|=O_p\left(n^{\frac{1}{2+\epsilon}}\right),$$

which means we can set

$$c_n=n^{\frac{1}{2+\epsilon}-\frac{1}{2}}=n^{-\frac{\epsilon}{2(2+\epsilon)}}=o(1).$$

This concludes our proof.

F.2 Remark 2

Part 1. To show the consistency of $\hat{\beta}$, we again need to find an integrable envelop function. Take an arbitrary β in the parameter space $\mathcal{B}$, then we consider the expectation

$$\mathbb{E}[D_i\ln\big(G(X_i'\beta)\big)]=\mathbb{E}[D_i\ln\big(G(X_i'\beta)\big)\mathbb{1}_{(X_i'\beta\leq-c)}]+\mathbb{E}[D_i\ln\big(G(X_i'\beta)\big)\mathbb{1}_{(X_i'\beta>-c)}],$$

for some c positive and large enough. Note that the second term is bounded from below by a finite constant which does not explicitly depend on the choice of β. As for the first term, we further employ a well-known bound on the normal tail probability (Proposition 2.1.2 in Vershynin 2018)

$$\begin{aligned}\mathbb{E}[D_i \ln\big(G(X_i'\beta)\big)\mathbb{1}_{(X_i'\beta\leq -c)}] &\geq \mathbb{E}\left[D_i \ln\left[\frac{1}{2X_i'\beta}g(X_i'\beta)\right]\mathbb{1}_{(X_i'\beta\geq c)}\right] \\ &= -\mathbb{E}\left[D_i \ln(2X_i'\beta)\mathbb{1}_{(X_i'\beta\geq c)}\right] - \mathbb{E}\left[D_i|X_i'\beta|^2\mathbb{1}_{(X_i'\beta\geq c)}\right],\end{aligned}$$

where $g(\cdot) = G^{(1)}(\cdot)$ is the standard normal density function. The calculation for the other term, $\mathbb{E}[(1-D_i)\ln\big(1-G(X_i'\beta)\big)]$ is analogous. From the above, it is clear that one can find an integrable envelop function as the parameter space is compact.

Next we consider the score

$$s_i(\beta) = \left[\frac{D_i}{G(X_i'\beta)} - \frac{1-D_i}{1-G(X_i'\beta)}\right] g(X_i'\beta)X_i.$$

To bound its second moment (at the true parameter β_0), it suffices to consider, for example,

$$\begin{aligned}&\mathbb{E}\left[\frac{D_i}{G(X_i'\beta_0)^2}g(X_i'\beta_0)^2\|X_i\|^2\right] \\ =&\mathbb{E}\left[\frac{1}{G(X_i'\beta_0)}g(X_i'\beta_0)^2\|X_i\|^2\right] \\ =&\mathbb{E}\left[\frac{1}{G(X_i'\beta_0)}g(X_i'\beta_0)^2\|X_i\|^2\mathbb{1}_{(X_i'\beta_0\leq -c)}\right] + \mathbb{E}\left[\frac{1}{G(X_i'\beta_0)}g(X_i'\beta_0)^2\|X_i\|^2\mathbb{1}_{(X_i'\beta_0> -c)}\right].\end{aligned}$$

Up to a multiplicative constant, the second term is finite provided that X_i has a finite variance. The first term can be bounded by

$$\mathbb{E}\left[\frac{1}{G(X_i'\beta_0)}g(X_i'\beta_0)^2\|X_i\|^2\mathbb{1}_{(X_i'\beta_0\leq -c)}\right] \leq \mathbb{E}\left[\frac{2|X_i'\beta_0|}{g(X_i'\beta_0)}g(X_i'\beta_0)^2\|X_i\|^2\mathbb{1}_{(X_i'\beta_0\leq -c)}\right],$$

which is finite if X_i has a finite third moment.

Supplementary Material

To characterize the $\sqrt{n}$-consistency and asymptotic normality of $\hat{\beta}$, what remains is to show the convergence to the Hessian matrix. To be precise, let

$$H_i(\beta) = \Bigg[D_i \frac{G(X_i'\beta)G^{(2)}(X_i'\beta) - G^{(1)}(X_i'\beta)^2}{G(X_i'\beta)^2} - (1-D_i)\frac{\Big(1-G(X_i'\beta)\Big)G^{(2)}(X_i'\beta) + G^{(1)}(X_i'\beta)^2}{\Big(1-G(X_i'\beta)\Big)^2} \Bigg] X_i X_i',$$

and we would like to show

$$\mathbb{E}\left[\sup_{\|\beta-\beta_0\| \le \frac{\epsilon}{\sqrt{n}}} \|H_i(\beta)\| \right] < \infty$$

for all $\epsilon > 0$. Consider, for example, the first half of the expectation:

$$\mathbb{E}\left[\sup_{\|\beta-\beta_0\| \le \frac{\epsilon}{\sqrt{n}}} D_i \left| \frac{G(X_i'\beta)G^{(2)}(X_i'\beta) - G^{(1)}(X_i'\beta)^2}{G(X_i'\beta)^2} \right| |X_i|^2 \right].$$

Again it suffices to restrict to the case where $X_i'\beta$ is sufficiently small. Then we can bound the denominator by

$$G(X_i'\beta) \ge \frac{1}{2|X_i'\beta|} g(X_i'\beta).$$

As a result, finiteness of the previous follows from a finite fourth moment of X_i. This concludes our proof of the $\sqrt{n}$-consistency and asymptotic normality of $\hat{\beta}$.

Part 2. Now we prove Assumption 3(ii) with an explicit order c_n. To start, observe

$$\begin{aligned}
&\left\| \frac{G(X_i'\beta_0)}{G(X_i'\beta)^2} \frac{\partial G(X_i'\beta)}{\partial \beta} \right\| \le \frac{G(X_i'\beta_0)}{G(X_i'\beta)^2} g(X_i'\beta)(1+\|X_i\|) \\
&\le \mathbb{1}_{(X_i'\beta \ge -2)} G(-2)^2 G(X_i'\beta_0) g(X_i'\beta)(1+\|X_i\|) + \mathbb{1}_{(X_i'\beta < -2)} \frac{G(X_i'\beta_0)}{G(X_i'\beta)^2} g(X_i'\beta)(1+\|X_i\|) \\
&\le \mathbb{1}_{(X_i'\beta \ge -2)} G(-2)^2 G(X_i'\beta_0) g(X_i'\beta)(1+\|X_i\|) + \mathbb{1}_{(X_i'\beta < -2)} \frac{g(X_i'\beta_0)}{g(X_i'\beta)} \left(\frac{|X_i'\beta|^3}{|X_i'\beta|^2 - 1} \right)^2 (1+\|X_i\|).
\end{aligned}$$

The first term above is easily bounded by

$$\mathbb{1}_{(X_i'\beta \geq -2)} G(-2)^2 G(X_i'\beta_0) g(X_i'\beta)(1 + \|X_i\|) \leq G(-2)^2 g(0)(1 + \|X_i\|),$$

and the second term can be bounded by

$$\mathbb{1}_{(X_i'\beta < -2)} \frac{g(X_i'\beta_0)}{g(X_i'\beta)} \left(\frac{|X_i'\beta|^3}{|X_i'\beta|^2 - 1} \right)^2 (1 + \|X_i\|) \leq \frac{16}{9} e^{\frac{1}{2}(1+\|X_i\|)^2 \|\beta - \beta_0\| \|\beta + \beta_0\|} (1 + \|X_i\|)^3.$$

Now assume $\|X_i\|$ has a finite $6 + \epsilon$ moment for some $\epsilon > 0$. Then

$$\max_{1 \leq i \leq n} \|X_i\| = O_p(n^{\frac{1}{6+\epsilon}}),$$

which implies that

$$\max_{1 \leq i \leq n} \sup_{\|\beta - \beta_0\| \leq \frac{c}{\sqrt{n}}} \left\| \frac{G(X_i'\beta_0)}{G(X_i'\beta)^2} \frac{\partial G(X_i'\beta)}{\partial \beta} \right\| = O_p(n^{\frac{3}{6+\epsilon}}).$$

Hence, we can set c_n to be

$$c_n = n^{\frac{3}{6+\epsilon} - \frac{1}{2}} = n^{-\frac{\epsilon}{2(6+\epsilon)}}.$$

This concludes our proof.

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